

AMTH 205: Numerical Methods

Numerical Integration

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Numerical Integration

Numerical Integration: The process of computing the value of a definite integral from a set of numerical values of the integrand is called numerical integration.

i.e., numerical integration is a technique calculating the numerical value of a definite integral.

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Fundamental Theorem of Calculus

If f is continuous on an interval $[a,b]$,

F is antiderivative of f (i.e., $F'(x) = f(x)$)

$$\int_a^b f(x)dx = F(b) - F(a)$$

There is no antiderivative for: e^{x^2}

No closed form solution for: $\int_a^b e^{x^2} dx$

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The Area Under the Curve

Integration

The process of measuring the area under a curve.

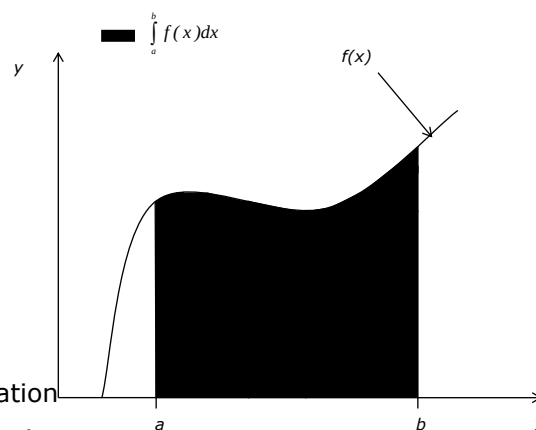
$$A = \int_a^b f(x)dx$$

Where:

$f(x)$ is the integrand

a = lower limit of integration

b = upper limit of integration



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Quadrature:

- To evaluate the above integral, we replace $y = f(x)$ by an approximate interpolating polynomial function $\phi(x)$ and get an approximate value of the given integral.
- **Quadrature:** When applied to the integration of a function of a single variable, the process is called quadrature.

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General quadrature formula

- Assume that $(n+1)$ points $(x_i, y_i), i = 0, 1, 2, \dots, n$ for any function $y = f(x)$ are given where f is not known explicitly. It is required to compute the values of the integral

$$I = \int_a^b y \, dx \dots\dots\dots(i)$$

- The values of x being equally spaced, i.e., $x_n = x_0 + nh$
- So (i) implies

$$I = \int_{x_0}^{x_n} y \, dx \dots\dots\dots(ii)$$

- But from the forward difference formula we know that,

$$f(x) = y = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots\dots\dots$$

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General quadrature formula

□ Therefore (ii) implies,

$$I = \int_{x_0}^{x_n} [y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots] dx$$

.....(iii)

Again, we know that, $\frac{x-x_0}{h} = p \Rightarrow x-x_0 = ph \Rightarrow dx = hdp$

Therefore (iii) implies

$$I = h \int_0^n [y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots] dp$$

$$I = h \left[p y_0 + \frac{p^2}{2} \Delta y_0 + \frac{\left(\frac{p^3}{3} - \frac{p^2}{2}\right)}{2!} \Delta^2 y_0 + \frac{\left(\frac{p^4}{4} - \frac{3p^3}{3} + p^2\right)}{3!} \Delta^3 y_0 + \dots \right]_0^n$$

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General quadrature formula

This implies,

$$I = h \left[n y_0 + \frac{n^2}{2} \Delta y_0 + \frac{1}{6} \frac{(2n^3 - 3n^2)}{2!} \Delta^2 y_0 + \frac{(n^4 - 4n^3 + 4n^2)}{24} \Delta^3 y_0 + \dots \right]$$

$$I = \int_{x_0}^{x_n} y dx = nh \left[y_0 + \frac{n}{2} \Delta y_0 + \frac{n(2n-3)}{12} \Delta^2 y_0 + \frac{n(n-2)^2}{24} \Delta^3 y_0 + \dots \right]$$

which is known as the general quadrature formula of integration. From this formula, we can obtain the following integration formula by replacing n by 1, 2, 3, and 6.

Trapezoidal rule is obtained for n = 1

Simpson's 1/3 rule is obtained for n = 2

Simpson's 3/8 rule is obtained for n = 3

Weddle's rule is obtained for n = 6

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Trapezoidal Rule

Trapezoidal Rule

- ▣ Trapezoidal rule is obtained by replacing n by 1.

$$I_1 = \int_{x_0}^{x_1} y dx = h \left[y_0 + \frac{1}{2} \Delta y_0 + 0 + \dots \right] = h \left[y_0 + \frac{1}{2} (y_1 - y_0) \right] = \frac{h}{2} (y_1 + y_0)$$

- ▣ Similarly

$$I_2 = \int_{x_1}^{x_2} y dx = \frac{h}{2} (y_2 + y_1)$$

$$I_3 = \int_{x_2}^{x_3} y dx = \frac{h}{2} (y_3 + y_2)$$

$$I_n = \int_{x_{n-1}}^{x_n} y dx = \frac{h}{2} (y_n + y_{n-1})$$

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Trapezoidal rule

- ▣ Adding the above integrals, we get

$$I = \int_{x_0}^{x_n} y dx = \frac{h}{2} (y_0 + 2(y_1 + y_2 + y_3 + \dots + y_{n-1}) + y_n)$$

which is known as the Trapezoidal rule.

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Simpson's 1/3 rule

- Simpson's 1/3 rule is obtained by replacing n by 2.

$$I_1 = \int_{x_0}^{x_2} y dx = 2h \left[y_0 + \frac{2}{2} \Delta y_0 + \frac{2(2*2-3)}{12} \Delta^2 y_0 + \dots \dots \dots \right] = 2h \left[y_0 + (y_1 - y_0) + \frac{1}{6} (y_2 - 2y_1 + y_0) \right]$$

$$= \frac{h}{3} (y_2 + 4y_1 + y_0)$$

- Similarly

$$I_2 = \int_{x_2}^{x_4} y dx = \frac{h}{3} (y_4 + 4y_3 + y_2)$$

$$I_3 = \int_{x_4}^{x_6} y dx = \frac{h}{3} (y_6 + 4y_5 + y_4)$$

$$I_{\frac{n}{2}} = \int_{x_{n-2}}^{x_n} y dx = \frac{h}{3} (y_n + 4y_{n-1} + y_{n-2})$$

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Simpson's 1/3 rule

- Adding the above integrals, we get

$$I = \int_{x_0}^{x_n} y dx = \frac{h}{3} (y_0 + 4(y_1 + y_3 + \dots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \dots + y_{n-2}) + y_n)$$

which is known as the Simpson's 1/3 rule.

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Simpson's 3/8 rule

- Simpson's 3/8 rule is obtained by replacing n by 3.

$$\begin{aligned} I_1 = \int_{x_0}^{x_3} y dx &= 3h \left[y_0 + \frac{3}{2} \Delta y_0 + \frac{3(2 \cdot 3 - 3)}{12} \Delta^2 y_0 + \frac{3(3 - 2)^2}{24} \Delta^3 y_0 + \dots \right] \\ &= 3h \left[y_0 + \frac{3}{2} (y_1 - y_0) + \frac{3}{4} (y_2 - 2y_1 + y_0) + \frac{1}{8} (y_3 - 3y_2 + 3y_1 - y_0) \right] \\ &= \frac{3h}{8} (y_0 + 3y_1 + 3y_2 + y_3) \end{aligned}$$

Similarly

$$I_2 = \int_{x_3}^{x_6} y dx = \frac{3h}{8} (y_3 + 3y_4 + 3y_5 + y_6)$$

$$I_3 = \int_{x_6}^{x_9} y dx = \frac{3h}{8} (y_6 + 3y_7 + 3y_8 + y_9)$$

$$I_{\frac{n}{3}} = \int_{x_{n-3}}^{x_n} y dx = \frac{3h}{8} (y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n)$$

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Simpson's 1/3 rule

- Adding the above integrals, we get

$$I = \int_{x_0}^{x_n} y dx = \frac{3h}{8} (y_0 + 3(y_1 + y_2 + y_4 + \dots + y_{n-2}) + 2(y_3 + y_6 + y_9 + \dots + y_{n-3}) + y_n)$$

which is known as the Simpson's 3/8 rule.

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Weddle's rule

- Weddle's rule is obtained by replacing n by 6.

$$I = \int_{x_0}^{x_6} y \, dx = \frac{3h}{10} (y_0 + 5(y_1 + y_5) + 2(y_2 + y_4) + 6y_3 + y_6)$$

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Newton-Cotes Methods

- In **Newton-Cote Methods**, the function is approximated by a **polynomial of order n** .
- Computing the integral of a polynomial is easy.

$$\int_a^b f(x) \, dx \approx \int_a^b (a_0 + a_1x + \dots + a_nx^n) \, dx$$

$$\int_a^b f(x) \, dx \approx a_0(b-a) + a_1 \frac{(b^2 - a^2)}{2} + \dots + a_n \frac{(b^{n+1} - a^{n+1})}{n+1}$$

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Newton-Cotes Methods

- **Trapezoid Method** (First Order Polynomials are used)

$$\int_a^b f(x)dx \approx \int_a^b (a_0 + a_1x)dx$$

- **Simpson 1/3 Rule** (Second Order Polynomials are used)

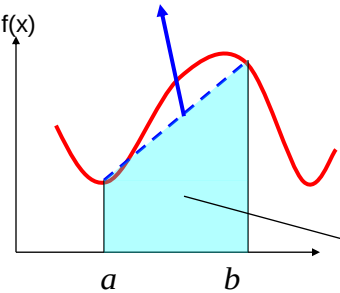
$$\int_a^b f(x)dx \approx \int_a^b (a_0 + a_1x + a_2x^2)dx$$

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Trapezoid Method

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Trapezoid Method



$$I = \int_a^b f(x) dx$$

$$I \approx \int_a^b \left(f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right) dx$$

$$= \left(f(a) - a \frac{f(b) - f(a)}{b - a} \right) x \Big|_a^b$$

$$+ \frac{f(b) - f(a)}{b - a} \frac{x^2}{2} \Big|_a^b$$

$$= (b - a) \frac{f(b) + f(a)}{2}$$

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Trapezoid Method

Derivation-One Interval

$$I = \int_a^b f(x) dx \approx \int_a^b \left(f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right) dx$$

$$I \approx \int_a^b \left(f(a) - a \frac{f(b) - f(a)}{b - a} + \frac{f(b) - f(a)}{b - a} x \right) dx$$

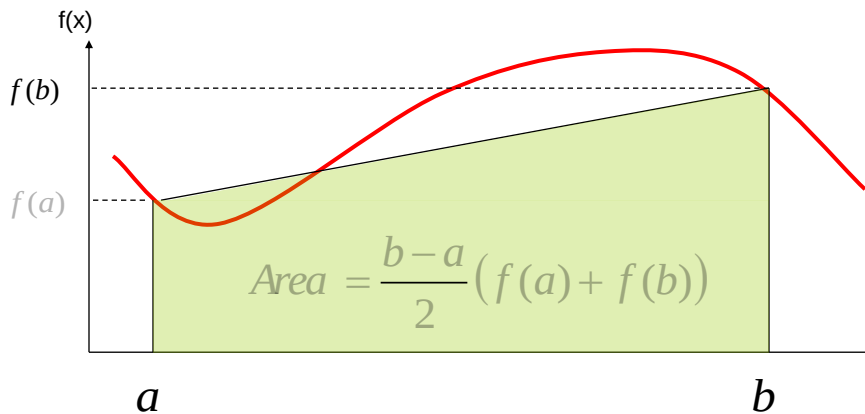
$$= \left(f(a) - a \frac{f(b) - f(a)}{b - a} \right) x \Big|_a^b + \frac{f(b) - f(a)}{b - a} \frac{x^2}{2} \Big|_a^b$$

$$= \left(f(a) - a \frac{f(b) - f(a)}{b - a} \right) (b - a) + \frac{f(b) - f(a)}{2(b - a)} (b^2 - a^2)$$

$$= (b - a) \frac{f(b) + f(a)}{2}$$

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Trapezoid Method

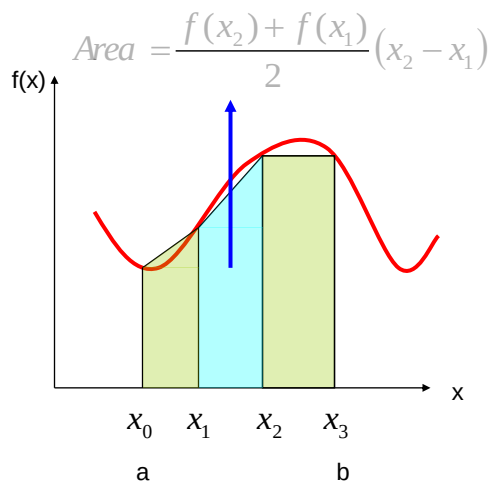


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Trapezoid Method

Multiple Application Rule

The interval $[a, b]$ is partitioned into n segments
 $a = x_0 \leq x_1 \leq x_2 \leq \dots \leq x_n = b$
 $\int_a^b f(x) dx = \text{sum of the areas of the trapezoids}$



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Trapezoid Method

General Formula and Special Case

If the interval is divided into n segments (not necessarily equal)

$$a = x_0 \leq x_1 \leq x_2 \leq \dots \leq x_n = b$$

$$\int_a^b f(x) dx \approx \sum_{i=0}^{n-1} \frac{1}{2} (x_{i+1} - x_i) (f(x_{i+1}) + f(x_i))$$

Special Case (Equally spaced base points)

$$x_{i+1} - x_i = h \quad \text{for all } i$$

$$\int_a^b f(x) dx \approx h \left[\frac{1}{2} [f(x_0) + f(x_n)] + \sum_{i=1}^{n-1} f(x_i) \right]$$

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Example by Trapezoidal rule

Using Trapezoidal rule compute the integral where the table for the values of f is given below:

X	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Y	1.00000	1.01005	1.04081	1.09417	1.17351	1.28402	1.43332	1.63231	1.89648	2.2479	2.71828

Ans 1.467171

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Example

$$\begin{aligned} I &\approx \frac{b-a}{2n} \left[f(a) + 2 \left\{ \sum_{i=1}^{n-1} f(a+ih) \right\} + f(b) \right] \\ I &\approx \frac{-108-80}{2(2)} \left[f(80) + 2 \left\{ \sum_{i=1}^{2-1} f(a+ih) \right\} + f(-108) \right] \\ &\approx \frac{-188}{4} [f(80) + 2f(80+1 \times (-94)) + f(-108)] \\ &\approx \frac{-188}{4} [f(80) + 2f(-14) + f(-108)] \\ &\approx -47 [7.9519 \times 10^{-5} + 2(7.3262 \times 10^{-5}) + 6.4322 \times 10^{-5}] \\ &\approx -0.013647 \text{ in} \end{aligned}$$

b) The exact value of the above integral is

$$\begin{aligned} \Delta D &= 12.363 \int_{80}^{-108} (-1.2278 \times 10^{-11} T^2 + 6.1946 \times 10^{-9} T + 6.015 \times 10^{-6}) dT \\ &= -0.013689 \text{ in} \end{aligned}$$

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Example

Values obtained using multiple-segment Trapezoidal rule

n	Value	E_t	$ \epsilon_t \%$	$ \epsilon_a \%$
1	-0.013521	-0.00016810	1.2280	---
2	-0.013647	-4.2026×10^{-5}	0.30700	0.92328
3	-0.013670	-1.8678×10^{-5}	0.13644	0.16825
4	-0.013679	-1.0506×10^{-5}	0.076750	0.059740
5	-0.013682	-6.7241×10^{-6}	0.049120	0.027644
6	-0.013684	-4.6695×10^{-6}	0.034111	0.015014
7	-0.013686	-3.4307×10^{-6}	0.025061	0.0090522
8	-0.013687	-2.6266×10^{-6}	0.019188	0.0058749

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Example

In an attempt to understand the mechanism of the depolarization process in a fuel cell, an electro-kinetic model for mixed oxygen-methanol current on platinum was developed in the laboratory at FAMU. A very simplified model of the reaction developed suggests a functional relation in an integral form. To find the time required for 50% of the oxygen to be consumed, the time, $T(s)$ is given by

$$T = -\int_{1.22 \times 10^{-6}}^{0.61 \times 10^{-6}} \left(\frac{6.73x + 4.3025 \times 10^{-7}}{2.316 \times 10^{-11} x} \right) dx$$

- Use single segment Trapezoidal rule to find the time required for 50% of the oxygen to be consumed.
- Find the true error, E_t , for part (a).
- Find the absolute relative true error, $|\epsilon_t|$, for part (a).

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Examples

Integrate $f(x) = x^3$ between $x=1$ and $x=2$.

$$\int_1^2 x^3 dx = \frac{1}{4} x^4 \Big|_1^2 = \frac{1}{4} (2^4 - 1^4) = 3.75$$

Using 4 subdivisions for the numerical integration: $\Delta x = \frac{2-1}{4} = 0.25$

Rectangular rule:

i	x_i^*	$f(x_i^*)$
1	1.125	1.42
2	1.375	2.60
3	1.625	4.29
4	1.875	6.59

$$\begin{aligned} \int_1^2 x^3 dx &= \Delta x [f(1.125) + f(1.375) + f(1.625) + f(1.875)] \\ &= 0.25(14.9) = 3.725 \end{aligned}$$

Trapezoidal Rule

i	x_i	$f(x_i)$
	1	1
1	1.25	1.95
2	1.5	3.38
3	1.75	5.36
	2	8

$$\int_1^2 x^3 dx$$

$$= \Delta x \left[\frac{1}{2} f(1) + f(1.25) + f(1.5) + f(1.75) + \frac{1}{2} f(2) \right]$$

$$= 0.25(15.19) = 3.80$$

Trapezoidal Rule

i	x_i	$f(x_i)$
	1	1
1	1.25	1.95
2	1.5	3.38
3	1.75	5.36
	2	8

$$\int_1^2 x^3 dx$$

$$= \Delta x \left[\frac{1}{2} f(1) + f(1.25) + f(1.5) + f(1.75) + \frac{1}{2} f(2) \right]$$

$$= 0.25(15.19) = 3.80$$

Simpson's Rule

$$\int_1^2 x^3 dx = \frac{\Delta x}{3} [f(1) + 4f(1.25) + 2f(1.5) + 4f(1.75) + f(2)]$$

$$= \frac{0.25}{3}(45) = 3.75 \Rightarrow \text{perfect estimation}$$

Example

Given a tabulated values of the velocity of an object.

Time (s)	0.0	1.0	2.0	3.0
Velocity (m/s)	0.0	10	12	14

Obtain an estimate of the distance traveled in the interval $[0,3]$.

Distance = integral of the velocity

$$\text{Distance} = \int_0^3 V(t) dt$$

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Example 1

The interval is divided into 3 subintervals

Base points are $\{0,1,2,3\}$

Time (s)	0.0	1.0	2.0	3.0
Velocity (m/s)	0.0	10	12	14

Trapezoid Method

$$h = x_{i+1} - x_i = 1$$

$$T = h \left[\sum_{i=1}^{n-1} f(x_i) + \frac{1}{2} (f(x_0) + f(x_n)) \right]$$

$$\text{Distance} = 1 \left[(10+12) + \frac{1}{2} (0+14) \right] = 29$$

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Example

A river is 80 feet wide. The depth d (in feet) of the river at a distance x from one bank is given by the following table:

x	0	10	20	30	40	50	60	70	80
d	0	4	7	9	12	15	14	8	3

Find approximately the area of the cross-section of the river using (i) Trapezoidal (ii) Simpson's 1/3 rule (iii) Simpson's 3/8 rule (iv) Weddle's rule.

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Error in Trapezoidal Rule

Let $y = f(x)$ be continuous and has continuous derivatives in $[x_0, x_1]$

Expanding y in a Taylor series around $x = x_0$ we obtain

$$\begin{aligned}
 y(x) &= y_0 + \frac{(x-x_0)}{1!} y'_0 + \frac{(x-x_0)^2}{2!} y''_0 + \dots (1) \\
 \int_{x_0}^{x_1} y dx &= \int_{x_0}^{x_1} \left(y_0 + \frac{(x-x_0)}{1!} y'_0 + \frac{(x-x_0)^2}{2!} y''_0 + \dots \right) dx \\
 &= \left(y_0 x + \frac{(x-x_0)^2}{2!} y'_0 + \frac{(x-x_0)^3}{3!} y''_0 + \dots \right)_{x_0}^{x_1} \\
 &= y_0 (x_1 - x_0) + \frac{(x_1 - x_0)^2}{2!} y'_0 + \frac{(x_1 - x_0)^3}{3!} y''_0 + \dots
 \end{aligned}$$

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Error in Trapezoidal Rule

$$\text{So } \int_{x_0}^{x_1} y dx = y_0 h + \frac{h^2}{2!} y'_0 + \frac{h^3}{3!} y''_0 + \dots (2)$$

$$\text{But } \int_{x_0}^{x_1} y dx = \frac{h}{2} (y_0 + y_1) \dots \dots \dots (3)$$

Putting $x = x_1$ in equation (1) we get

$$y_1 = y_0 + \frac{(x_1 - x_0)}{1!} y'_0 + \frac{(x_1 - x_0)^2}{2!} y''_0 +$$

$$\Rightarrow y_1 = y_0 + h y'_0 + \frac{h^2}{2!} y''_0 + \dots \dots (4)$$

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Error in Trapezoidal Rule

$$\text{Now } \frac{h}{2} (y_0 + y_1) = \frac{h}{2} \left(y_0 + y_0 + h y'_0 + \frac{h^2}{2!} y''_0 + \right)$$

$$= \frac{h}{2} \left(2y_0 + h y'_0 + \frac{h^2}{2!} y''_0 + \right) \dots \dots (5)$$

From equation (2) and (5) we obtain

$$\begin{aligned} \int_{x_0}^{x_1} y dx - \frac{h}{2} (y_0 + y_1) &= y_0 h + \frac{h^2}{2!} y'_0 + \frac{h^3}{3!} y''_0 + \dots - \frac{h}{2} \left(2y_0 + h y'_0 + \frac{h^2}{2!} y''_0 + \right) \\ &= -\frac{1}{12} h^3 y''_0 + \dots \end{aligned}$$

$$\text{Hence the local error in the interval } [x_0, x_1] = -\frac{1}{12} h^3 y''_0$$

i.e. Local error of T.R. is of order (h^3)

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Error in Trapezoidal Rule

Now, Global error in T.R. means the sum of all n local errors, which is

$$E = -\frac{1}{12} h^3 (y_0'' + y_1'' + y_2'' + \dots + y_n'')$$

$$|E| < \frac{nh^3}{12} M, \text{ where } M = \max \{|y_0''|, |y_1''|, |y_2''|, \dots, |y_n''|\}$$

$$\therefore |E| < \frac{h^2(b-a)}{12} M$$

Hence the global error in the Trapezoidal rule is of order (h^2) which is one less to the order of local error.

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Error in Simpson's Rule

Global error in Simpson's 1/3 rule is

$$\therefore |E| < \frac{h^4(b-a)}{180} M$$

Global error in Simpson's 3/8 rule is

$$\therefore |E| < \frac{h^4(b-a)}{80} M$$

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Estimating the Error

For Trapezoid Method

How many equally spaced intervals are needed to compute $\int_0^{\pi} \sin(x) dx$ to 5 decimal digit accuracy?

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Example

$$\int_0^{\pi} \sin(x) dx, \quad \text{find } h \text{ so that } |\text{error}| \leq \frac{1}{2} \times 10^{-5}$$

$$|\text{Error}| \leq \frac{b-a}{12} h^2 \max_{x \in [a,b]} |f''(x)|$$

$$b = \pi; \quad a = 0; \quad f'(x) = \cos(x); \quad f''(x) = -\sin(x)$$

$$|f''(x)| \leq 1 \Rightarrow |\text{Error}| \leq \frac{\pi}{12} h^2 \leq \frac{1}{2} \times 10^{-5}$$

$$\Rightarrow h^2 \leq \frac{6}{\pi} \times 10^{-5} \Rightarrow h \leq 0.00437$$

$$\Rightarrow n \geq \frac{(b-a)}{h} = \frac{\pi}{0.00437} = 719 \text{ intervals}$$

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Example

x	1.0	1.5	2.0	2.5	3.0
f(x)	2.1	3.2	3.4	2.8	2.7

Use Trapezoid method to compute: $\int_1^3 f(x)dx$

$$\text{Trapezoid } T(f, P) = \sum_{i=0}^{n-1} \frac{1}{2} (x_{i+1} - x_i) (f(x_{i+1}) + f(x_i))$$

Special Case: $h = x_{i+1} - x_i$ for all i ,

$$T(f, P) = h \left[\sum_{i=1}^{n-1} f(x_i) + \frac{1}{2} (f(x_0) + f(x_n)) \right]$$

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Example

x	1.0	1.5	2.0	2.5	3.0
f(x)	2.1	3.2	3.4	2.8	2.7

$$\begin{aligned} \int_1^3 f(x)dx &\approx h \left[\sum_{i=1}^{n-1} f(x_i) + \frac{1}{2} (f(x_0) + f(x_n)) \right] \\ &= 0.5 \left[3.2 + 3.4 + 2.8 + \frac{1}{2} (2.1 + 2.7) \right] \\ &= 5.9 \end{aligned}$$

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Motivation for Romberg Method

- Romberg Integration is an extrapolation formula of the Trapezoidal Rule for integration. It provides a better approximation of the integral by reducing the True Error.
- Trapezoid formula with a sub-interval h gives an error of the order $O(h^2)$.
- *We can combine two Trapezoid estimates with intervals h and $h/2$ to get a better estimate.*

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Romberg Integration

The method arises from a technique called Richardson extrapolation which can be used whenever the error $E(h)$ can be expanded in a series of the form

$$E(h) = c_1h + c_2h^2 + c_3h^3 + c_4h^4 + \dots$$

We'll illustrate Richardson's technique by applying it to the trapezoidal rule.

$$I = \int_a^b f(x)dx = T(h) + E(h), \quad \text{with } E(h) = O(h^2)$$

In fact we can show that if f can be expanded in a Taylor series on each subinterval then E can be expanded in a series of the form

$$E(h) = c_2h^2 + c_4h^4 + c_6h^6 + c_8h^8 + \dots$$

To implement the method we don't need to know the coefficients, we need only know that they exist. Assume that we have computed trapezoidal estimates of I for $h, h/2, h/4, h/8$, so we have $T(h), T(h/2), T(h/4), T(h/8)$.

Just looking at the first two we have

$$I = T(h) + E(h) = T(h) + c_2 h^2 + c_4 h^4 + c_6 h^6 + c_8 h^8 + \dots$$

$$\text{and } I = T\left(\frac{h}{2}\right) + E\left(\frac{h}{2}\right) = T\left(\frac{h}{2}\right) + c_2 \frac{h^2}{2^2} + c_4 \frac{h^4}{2^4} + c_6 \frac{h^6}{2^6} + c_8 \frac{h^8}{2^8} + \dots$$

We can eliminate the h^2 term by multiplying the second by 2^2 and subtracting the first giving

$$(2^2 - 1)I = 2^2 T\left(\frac{h}{2}\right) - T(h) + c'_4 \frac{h^4}{2^4} + c'_6 \frac{h^6}{2^6} + c'_8 \frac{h^8}{2^8} + \dots$$

Which we can rewrite as

$$(2^2 - 1)I = (2^2 - 1)T\left(\frac{h}{2}\right) + T\left(\frac{h}{2}\right) - T(h) + c'_4 \frac{h^4}{2^4} + c'_6 \frac{h^6}{2^6} + c'_8 \frac{h^8}{2^8} + \dots$$

Romberg Integration

The last equation is equivalent to

$$I = T\left(\frac{h}{2}\right) + \frac{T\left(\frac{h}{2}\right) - T(h)}{(2^2 - 1)} + k_4 \frac{h^4}{2^4} + k_6 \frac{h^6}{2^6} + k_8 \frac{h^8}{2^8} + \dots$$

Which is in the form

$$\begin{aligned} I &= \text{estimate} + \text{error} \\ &= T_2(h) + k_4 \frac{h^4}{2^4} + k_6 \frac{h^6}{2^6} + k_8 \frac{h^8}{2^8} + \dots \end{aligned}$$

$T_2(h)$ is an estimate for I with an error that is $O(h^4)$

We can repeat the process beginning with

$$I = T_2(h) + k_4 \frac{h^4}{2^4} + k_6 \frac{h^6}{2^6} + k_8 \frac{h^8}{2^8} + \dots$$

And get a new estimate with

$$I = T_3(h) + k'_6 \frac{h^6}{2^6} + k'_8 \frac{h^8}{2^8} + \dots$$

$$T_3(h) = T_2\left(\frac{h}{2}\right) + \frac{T_2\left(\frac{h}{2}\right) - T_2(h)}{(2^4 - 1)}$$

Continuing in this way we get a sequence of estimates for the integral I

$$T_1(h) = T(h), \quad T_j(h) = T_{j-1}\left(\frac{h}{2}\right) + \frac{T_{j-1}\left(\frac{h}{2}\right) - T_{j-1}(h)}{(2^{2(j-1)} - 1)}$$

Where for each estimate we have

$$I = T_j(h) + E_j(h), \quad \text{with } E_j(h) = O(h^{2j})$$

Romberg Integration -- An example

To see how Romberg integration works we organize our estimates into an array as follows

$$\begin{array}{cccc} T_1(h) & & & \\ T_1\left(\frac{h}{2}\right) & T_2(h) & & \\ T_1\left(\frac{h}{4}\right) & T_2\left(\frac{h}{2}\right) & T_3(h) & \\ T_1\left(\frac{h}{8}\right) & T_2\left(\frac{h}{4}\right) & T_3\left(\frac{h}{2}\right) & T_4(h) \end{array}$$

The accuracy increases as we move down and as we move to the right.

Next we perform Romberg integration for a familiar example.

$$\int_1^3 x^{3/2} dx$$

We place our trapezoidal estimates in the first column. The first estimate corresponds to a partition of $[1, 3]$ into 4 subintervals, an h value of $\frac{1}{2}$. Accuracy improves as we move down the column

$$\int_1^3 x^{3/2} dx = \frac{2}{5} x^{5/2} \Big|_1^3 = \frac{2}{5} (3^{5/2} - 1^{5/2}) = 5.835382907$$

5.858234			
5.841100	5.835388		
5.836813	5.835384	5.835384	
5.835740	5.835382	5.835382	5.835382

Romberg Integration -- An example

We can improve our Romberg result if we start with more precise (though quite inaccurate) trapezoidal estimates in the first column. The averaging method used to get the more accurate estimates involves the subtraction of nearly equal numbers which can lead to precision loss as the computation progresses. The distinction between precision and accuracy in computation is important in practical numerical analysis.

5.85823385920
5.84110042206 5.83538927635
5.83681259147 5.83538331460 5.83538291715
5.83574034752 5.83538293287 5.83538290742 5.83538290726

Note: The correct value to 12 significant digits is 5.83538290725

The values for Romberg’s Method of integration can be arranged in a triangular matrix.

	0	1	2	3	4	5	The best possible estimate in each row is the right most entry: $A_0, B_1, C_1, D_1, E_1, F_1$
0	A_0						
1	A_1	B_1					
2	A_2	B_2	C_1				
3	A_3	B_3	C_2	D_1			
4	A_4	B_4	C_3	D_2	E_1		
5	A_5	B_5	C_4	D_3	E_2	F_1	

Since we will run out of letters after 26 rows we express this with $R(i,j)$.

	0	1	2	3	4	5	The best possible estimate in each row is the right most entry: $R(i,i)$
0	$R(0,0)$						
1	$R(1,0)$	$R(1,1)$					
2	$R(2,0)$	$R(2,1)$	$R(2,2)$				
3	$R(3,0)$	$R(3,1)$	$R(3,2)$	$R(3,3)$			
4	$R(4,0)$	$R(4,1)$	$R(4,2)$	$R(4,3)$	$R(4,4)$		
5	$R(5,0)$	$R(5,1)$	$R(5,2)$	$R(5,3)$	$R(5,4)$	$R(5,5)$	

The $R(i,j)$ can be obtained in the following way:

$R(i,0)$ is always the trapezoid method applied with a partition size of 2^i .

$$R(i,j) = \frac{4^j R(i,j-1) - R(i-1,j-1)}{4^j - 1}$$

	0	1	2	3	4	5
0	$R(0,0)$					
1	$R(1,0)$	$R(1,1)$				
2	$R(2,0)$	$R(2,1)$	$R(2,2)$			
3	$R(3,0)$	$R(3,1)$	$R(3,2)$	$R(3,3)$		
4	$R(4,0)$	$R(4,1)$	$R(4,2)$	$R(4,3)$	$R(4,4)$	
5	$R(5,0)$	$R(5,1)$	$R(5,2)$	$R(5,3)$	$R(5,4)$	$R(5,5)$

For the rest of them a triangular pattern exists as how you fill in the other entries.

$$R(1,1) = \frac{4R(1,0) - R(0,0)}{4 - 1}$$

$$R(2,1) = \frac{4R(2,0) - R(1,0)}{4 - 1}$$

$$R(3,1) = \frac{4R(3,0) - R(2,0)}{4 - 1}$$

$$R(4,1) = \frac{4R(4,0) - R(3,0)}{4 - 1}$$

$$R(5,1) = \frac{4R(5,0) - R(4,0)}{4 - 1}$$

$$R(2,2) = \frac{4^2 R(2,1) - R(1,1)}{4^2 - 1}$$

$$R(3,2) = \frac{4^2 R(3,1) - R(2,1)}{4^2 - 1}$$

$$R(4,2) = \frac{4^2 R(4,1) - R(3,1)}{4^2 - 1}$$

$$R(5,2) = \frac{4^2 R(5,1) - R(4,1)}{4^2 - 1}$$

$$R(3,3) = \frac{4^3 R(3,2) - R(2,2)}{4^3 - 1}$$

$$R(4,3) = \frac{4^3 R(4,2) - R(3,2)}{4^3 - 1}$$

$$R(5,3) = \frac{4^3 R(5,2) - R(4,2)}{4^3 - 1}$$

Example: Apply Romberg Extrapolation to the data to the right to get the best possible estimate for:

$$\int_0^2 f(x) dx$$

$$R(0,0) = \frac{1}{2} \left(\frac{2-0}{1} \right) (0 + 16) = 16$$

$$R(1,0) = \frac{1}{2} \left(\frac{2-0}{2} \right) (0 + 2 \cdot 4 + 16) = 12$$

$$R(1,1) = \frac{4(12) - 16}{3} = \frac{32}{3}$$

$$R(2,0) = \frac{1}{2} \left(\frac{2-0}{4} \right) (0 + 2 \cdot 1 + 2 \cdot 4 + 2 \cdot 9 + 16) = 11$$

$$R(2,1) = \frac{4(11) - 12}{3} = \frac{32}{3}$$

$$R(2,2) = \frac{4^2 \left(\frac{32}{3} \right) - \frac{32}{3}}{4^2 - 1} = \frac{32}{3}$$

i	x_i	$y_i = f(x_i)$
0	0	0
1	$\frac{1}{2}$	1
2	1	4
3	$\frac{3}{2}$	9
4	2	16

The pattern that exists for the Romberg Entries is as follows:

	$j-1$	j
$i-1$	$R(i-1,j-1)$	
i	$R(i,j-1)$	$R(i,j)$

$$R(i,j) = \frac{4^j R(i,j-1) - R(i-1,j-1)}{4^j - 1}$$

Note: This only applies for $i > 0$ and $j > 0$.

Example

Estimate the integral to the right with 3 rows of the Romberg Method of integration.

$$\int_0^{2\pi} \frac{x^2}{\pi^2} \sin x dx \approx ?$$

$$R(0,0) = \frac{1}{2}(2\pi - 0)(0 + 0) = 0$$

$$R(1,0) = \frac{1}{2}\left(\frac{2\pi-0}{2}\right)(0 + 2 \cdot 0 + 0) = 0$$

$$R(2,0) = \frac{1}{2}\left(\frac{2\pi-0}{4}\right)(0 + 2 \cdot \frac{1}{4} + 2 \cdot 0 + 2 \cdot \frac{-9}{4}) = \frac{\pi}{4}(-4) = -\pi$$

Building the triangular matrix:

$$R(0,0) = 0$$

$$R(1,0) = 0 \quad R(1,1) = \frac{4 \cdot 0 - 0}{3} = 0$$

$$R(2,0) = -\pi \quad R(2,1) = \frac{4 \cdot (-\pi) - 0}{3} = \frac{-4\pi}{3} \quad R(2,2) = \frac{16 \cdot \left(\frac{-4\pi}{3}\right) - 0}{15} = \frac{-64\pi}{45}$$

Example 1

The vertical distance covered by a rocket from 8 to 30 seconds is given by

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

- Use Richardson's rule to find the distance covered. Use the 2-segment and 4-segment Trapezoidal rule results given in Table 1.
- Find the true error, E_t for part (a).
- Find the absolute relative true error, $|\epsilon_a|$ for part (a).

Solution

a) $I_2 = 11266m$ $I_4 = 11113m$

Using Richardson's extrapolation formula for Trapezoidal rule

$$TV \approx I_{2n} + \frac{I_{2n} - I_n}{3} \quad \text{and choosing } n=2,$$

$$\begin{aligned} TV &\approx I_4 + \frac{I_4 - I_2}{3} = 11113 + \frac{11113 - 11266}{3} \\ &= 11062m \end{aligned}$$

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Solution (cont.)

b) The exact value of the above integral is

$$\begin{aligned} x &= \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt \\ &= 11061 m \end{aligned}$$

Hence

$$\begin{aligned} E_t &= \text{True Value} - \text{Approximate Value} \\ &= 11061 - 11062 \\ &= -1 m \end{aligned}$$

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Solution (cont.)

c) The absolute relative true error $|\epsilon_t|$ would then be

$$|\epsilon_t| = \left| \frac{11061 - 11062}{11061} \right| \times 100$$
$$= 0.00904\%$$

Table 2 shows the Richardson’s extrapolation results using 1, 2, 4, 8 segments. Results are compared with those of Trapezoidal rule.

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Solution (cont.)

Table 2: The values obtained using Richardson’s extrapolation formula for Trapezoidal rule for

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

n	Trapezoidal Rule	$ \epsilon_t $ for Trapezoidal Rule	Richardson’s Extrapolation	$ \epsilon_t $ for Richardson’s Extrapolation
1	11868	7.296	--	--
2	11266	1.854	11065	0.03616
4	11113	0.4655	11062	0.009041
8	11074	0.1165	11061	0.0000

Table 2: Richardson’s Extrapolation Values

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Romberg Integration

Romberg integration is same as Richardson's extrapolation formula as given previously. However, Romberg used a recursive algorithm for the extrapolation. Recall

$$TV \approx I_{2n} + \frac{I_{2n} - I_n}{3}$$

This can alternately be written as

$$(I_{2n})_R = I_{2n} + \frac{I_{2n} - I_n}{3} = I_{2n} + \frac{I_{2n} - I_n}{4^{2-1} - 1}$$

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Romberg Integration

Note that the variable TV is replaced by $(I_{2n})_R$ as the value obtained using Richardson's extrapolation formula. Note also that the sign $\approx$ is replaced by $=$ sign.

Hence the estimate of the true value now is

$$TV \approx (I_{2n})_R + Ch^4$$

Where Ch^4 is an approximation of the true error.

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Romberg Integration

Determine another integral value with further halving the step size (doubling the number of segments),

$$(I_{4n})_R = I_{4n} + \frac{I_{4n} - I_{2n}}{3}$$

It follows from the two previous expressions that the true value TV can be written as

$$\begin{aligned} TV &\approx (I_{4n})_R + \frac{(I_{4n})_R - (I_{2n})_R}{15} \\ &= I_{4n} + \frac{(I_{4n})_R - (I_{2n})_R}{4^{3-1} - 1} \end{aligned}$$

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Romberg Integration

A general expression for Romberg integration can be written as

$$I_{k,j} = I_{k-1,j+1} + \frac{I_{k-1,j+1} - I_{k-1,j}}{4^{k-1} - 1}, k \geq 2$$

The index k represents the order of extrapolation.

$k=1$ represents the values obtained from the regular

Trapezoidal rule, $k=2$ represents values obtained using the true estimate as $O(h^2)$. The index j represents the more and less accurate estimate of the integral.

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Example 2

The vertical distance covered by a rocket from $t = 8$ to $t = 30$ seconds is given by

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

Use Romberg's rule to find the distance covered. Use the 1, 2, 4, and 8-segment Trapezoidal rule results as given in the Table 1.

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Solution

From Table 1, the needed values from original Trapezoidal rule are

$$I_{1,1} = 11868 \quad I_{1,2} = 11266$$

$$I_{1,3} = 11113 \quad I_{1,4} = 11074$$

where the above four values correspond to using 1, 2, 4 and 8 segment Trapezoidal rule, respectively.

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Solution (cont.)

To get the first order extrapolation values,

$$\begin{aligned} I_{2,1} &= I_{1,2} + \frac{I_{1,2} - I_{1,1}}{3} \\ &= 11266 + \frac{11266 - 11868}{3} \\ &= 11065 \end{aligned}$$

Similarly,

$$\begin{aligned} I_{2,2} &= I_{1,3} + \frac{I_{1,3} - I_{1,2}}{3} & I_{2,3} &= I_{1,4} + \frac{I_{1,4} - I_{1,3}}{3} \\ &= 11113 + \frac{11113 - 11266}{3} & &= 11074 + \frac{11074 - 11113}{3} \\ &= 11062 & &= 11061 \end{aligned}$$

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Solution (cont.)

For the second order extrapolation values,

$$\begin{aligned} I_{3,1} &= I_{2,2} + \frac{I_{2,2} - I_{2,1}}{15} \\ &= 11062 + \frac{11062 - 11065}{15} \\ &= 11062 \end{aligned}$$

Similarly,

$$\begin{aligned} I_{3,2} &= I_{2,3} + \frac{I_{2,3} - I_{2,2}}{15} \\ &= 11061 + \frac{11061 - 11062}{15} \\ &= 11061 \end{aligned}$$

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Solution (cont.)

For the third order extrapolation values,

$$\begin{aligned} I_{4,1} &= I_{3,2} + \frac{I_{3,2} - I_{3,1}}{63} \\ &= 11061 + \frac{11061 - 11062}{63} \\ &= 11061m \end{aligned}$$

Table 3 shows these increased correct values in a tree graph.

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Solution (cont.)

Table 3: Improved estimates of the integral value using Romberg Integration

		First Order	Second Order	Third Order
1-segment	11868			
2-segment	1126			
4-segment	11113			
8-segment	11074			
		11065	11062	11061
		11062		
		11061	11061	

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Example 2

A company advertises that every roll of toilet paper has at least 250 sheets. The probability that there are 250 or more sheets in the toilet paper is given by

$$P(y \geq 250) = \int_{250}^{\infty} 0.3515 e^{-0.3881(y-252.2)^2} dy$$

Table 1 Values obtained using Trapezoidal rule.

Approximating the above integral as

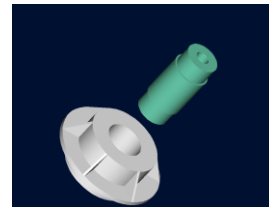
$$P(y \geq 250) = \int_{250}^{270} 0.3515 e^{-0.3881(y-252.2)^2} dy$$

n	Trapezoidal Rule
1	0.53721
2	0.26861
4	0.21814
8	0.95767

- Use Romberg's rule to find the probability. Use the 1, 2, 4, and 8-segment Trapezoidal rule results given in Table 1.
- Find the true error for part (a).
- Find the absolute relative true error for part (a).

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Example 3



A trunnion of diameter 12.363" has to be cooled from a room temperature of 80°F before it is shrink fit into a steel hub (Figure 1). The equation that gives the diametric contraction, in inches of the trunnion in dry-ice/alcohol (boiling temperature is -108°F) is given by:

$$\Delta D = 12.363 \int_{80}^{-108} (-1.2278 \times 10^{-11} T^2 + 6.1946 \times 10^{-9} T + 6.015 \times 10^{-6}) dT$$

Use Romberg's rule to find the contraction.

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Romberg Method

Estimates using Trapezoid method intervals of size $h, h/2, h/4, h/8 \dots$

are combined to improve the approximation of $\int_a^b f(x) dx$

First column is obtained using Trapezoid Method

R(0,0)			
R(1,0)	R(1,1)		
R(2,0)	R(2,1)	R(2,2)	
R(3,0)	R(3,1)	R(3,2)	R(3,3)

The other elements are obtained using the Romberg Method

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First Column Recursive Trapezoid Method

$$R(0,0) = \frac{b-a}{2} [f(a) + f(b)]$$

$$R(n,0) = \frac{1}{2} R(n-1,0) + h \left[\sum_{k=1}^{2^{(n-1)}} f(a + (2k-1)h) \right]$$

$$h = \frac{b-a}{2^n}$$

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Derivation of Romberg Method

$$\int_a^b f(x)dx = R(n-1,0) + O(h^2) \quad \text{Trapezoid method with } h = \frac{b-a}{2^{n-1}}$$

$$\int_a^b f(x)dx = R(n-1,0) + a_2h^2 + a_4h^4 + a_6h^6 + \dots \quad (eq1)$$

More accurate estimate is obtained by $R(n,0)$

$$\int_a^b f(x)dx = R(n,0) + \frac{1}{4}a_2h^2 + \frac{1}{16}a_4h^4 + \frac{1}{64}a_6h^6 + \dots \quad (eq2)$$

$eq1 - 4 * eq2$ gives

$$\int_a^b f(x)dx = \frac{1}{3}[4 \times R(n,0) - R(n-1,0)] + b_4h^4 + b_6h^6 + \dots$$

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Romberg Method

$$R(0,0) = \frac{b-a}{2} [f(a) + f(b)]$$

$$h = \frac{b-a}{2^n},$$

R(0,0)			
R(1,0)	R(1,1)		
R(2,0)	R(2,1)	R(2,2)	
R(3,0)	R(3,1)	R(3,2)	R(3,3)

$$R(n,0) = \frac{1}{2}R(n-1,0) + h \left[\sum_{k=1}^{2^{(n-1)}} f(a + (2k-1)h) \right]$$

$$R(n,m) = \frac{1}{4^m - 1} \left[4^m \times R(n,m-1) - R(n-1,m-1) \right] \quad n \geq 1, \quad m \geq 1$$

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Property of Romberg Method

Theorem

$$\int_a^b f(x)dx = R(n,m) + O(h^{2m+2})$$

R(0,0)			
R(1,0)	R(1,1)		
R(2,0)	R(2,1)	R(2,2)	
R(3,0)	R(3,1)	R(3,2)	R(3,3)

Error Level

$O(h^2)$
 $O(h^4)$
 $O(h^6)$
 $O(h^8)$

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Example

Compute $\int_0^1 x^2dx$

0.5	
3/8	1/3

$$h = 1, R(0,0) = \frac{b-a}{2} [f(a) + f(b)] = \frac{1}{2} [0 + 1] = 0.5$$

$$h = \frac{1}{2}, R(1,0) = \frac{1}{2} R(0,0) + h (f(a+h)) = \frac{1}{2} \left(\frac{1}{2}\right) + \frac{1}{2} \left(\frac{1}{4}\right) = \frac{3}{8}$$

$$R(n,m) = \frac{1}{4^m - 1} \left[4^m \times R(n,m-1) - R(n-1,m-1) \right] \text{ for } n \geq 1, m \geq 1$$

$$R(1,1) = \frac{1}{4^1 - 1} \left[4 \times R(1,0) - R(0,0) \right] = \frac{1}{3} \left[4 \times \frac{3}{8} - \frac{1}{2} \right] = \frac{1}{3}$$

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Example (cont.)

0.5		
3/8	1/3	
11/32	1/3	1/3

$$h = \frac{1}{4}, R(2,0) = \frac{1}{2}R(1,0) + h(f(a+h) + f(a+3h)) = \frac{1}{2}\left(\frac{3}{8}\right) + \frac{1}{4}\left(\frac{1}{16} + \frac{9}{16}\right) = \frac{11}{32}$$

$$R(n,m) = \frac{1}{4^m - 1} \left[4^m \times R(n, m-1) - R(n-1, m-1) \right]$$

$$R(2,1) = \frac{1}{3} \left[4 \times R(2,0) - R(1,0) \right] = \frac{1}{3} \left[4 \times \frac{11}{32} - \frac{3}{8} \right] = \frac{1}{3}$$

$$R(2,2) = \frac{1}{4^2 - 1} \left[4^2 \times R(2,1) - R(1,1) \right] = \frac{1}{15} \left[\frac{16}{3} - \frac{1}{3} \right] = \frac{1}{3}$$

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When do we stop?

STOP if

$$|R(n,n) - R(n,n-1)| \leq \varepsilon$$

or

After a given number of steps,

for example, STOP at R(4,4)

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- All electrical components, especially off-the-shelf components do not match their nominal value. Variations in materials and manufacturing as well as operating conditions can affect their value. Suppose a circuit is designed such that it requires a specific component value, how confident can we be that the variation in the component value will result in acceptable circuit behavior? To solve this problem a probability density function is needed to be integrated to determine the confidence interval. For an oscillator to have its frequency within 5% of the target of 1 kHz, the likelihood of this happening can then be determined by finding the total area under the normal distribution for the range in question:

$$(1 - \alpha) = \int_{-2.15}^{2.9} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

- Use Romberg's rule to find the frequency. Use the 1, 2, 4, and 8-segment Trapezoidal rule results as given.

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Gauss Quadrature

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Motivation

Trapezoid Method :

$$\int_a^b f(x)dx \approx h \left[\sum_{i=1}^{n-1} f(x_i) + \frac{1}{2}(f(x_0) + f(x_n)) \right]$$

It can be expressed as :

$$\int_a^b f(x)dx \approx \sum_{i=0}^n c_i f(x_i)$$

$$\text{where } c_i = \begin{cases} h & i = 1, 2, \dots, n-1 \\ 0.5h & i = 0 \text{ and } n \end{cases}$$

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General Integration Formula

$$\int_a^b f(x)dx \approx \sum_{i=0}^n c_i f(x_i)$$

c_i : Weights x_i : Nodes

Problem:

How do we select c_i and x_i so that the formula gives a good approximation of the integral?

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Lagrange Interpolation

$$\int_a^b f(x)dx \approx \int_a^b P_n(x)dx$$

where $P_n(x)$ is a polynomial that interpolates $f(x)$ at the nodes: $x_0, x_1, \dots, x_n$

$$\int_a^b f(x)dx \approx \int_a^b P_n(x)dx = \int_a^b \left(\sum_{i=0}^n \ell_i(x) f(x_i) \right) dx$$

$$\Rightarrow \int_a^b f(x)dx \approx \sum_{i=0}^n c_i f(x_i) \quad \text{where } c_i = \int_a^b \ell_i(x) dx$$

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Example

- ▣ Determine the Gauss Quadrature Formula of $\int_{-2}^2 f(x)dx$

If the nodes are given as $(-1, 0, 1)$

- ▣ Solution: First we need to find $l_0(x), l_1(x), l_2(x)$
- ▣ Then compute:

$$c_0 = \int_{-2}^2 l_0(x)dx, \quad c_1 = \int_{-2}^2 l_1(x)dx, \quad c_2 = \int_{-2}^2 l_2(x)dx$$

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Solution

$$l_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{x(x-1)}{2}$$

$$l_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = -(x+1)(x-1)$$

$$l_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{x(x+1)}{2}$$

$$c_0 = \int_{-2}^2 \frac{x(x-1)}{2} dx = \frac{8}{3}, \quad c_1 = \int_{-2}^2 -(x+1)(x-1) dx = -\frac{4}{3}, \quad c_2 = \int_{-2}^2 \frac{x(x+1)}{2} dx = \frac{8}{3}$$

The Gauss Quadrature Formula for $\int_{-2}^2 f(x) dx = \frac{8}{3} f(-1) - \frac{4}{3} f(0) + \frac{8}{3} f(1)$

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Using the Gauss Quadrature Formula

Case 1: Let $f(x) = x^2$

The exact value for $\int_{-2}^2 f(x) dx = \int_{-2}^2 x^2 dx = \frac{16}{3}$

The Gauss Quadrature Formula $= \frac{8}{3} f(-1) - \frac{4}{3} f(0) + \frac{8}{3} f(1)$

$= \frac{8}{3}(-1)^2 - \frac{4}{3}(0)^2 + \frac{8}{3}(1)^2 = \frac{16}{3}$, which is the same exact answer

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Improper Integrals

Methods discussed earlier cannot be used directly to approximate improper integrals (one of the limits is ∞ or $-\infty$)

$\Rightarrow$ Use a transformation like the following

$$\int_a^b f(x) dx = \int_{\frac{1}{b}}^{\frac{1}{a}} \frac{1}{t^2} f\left(\frac{1}{t}\right) dt, \quad (\text{assuming } ab > 0)$$

and apply the method on the new function.

Example:
$$\int_1^{\infty} \frac{1}{x^2} dx = \int_0^1 \frac{1}{t^2} \left(\frac{1}{\left(\frac{1}{t}\right)^2} \right) dt$$

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